

y3hbxpyse

May 7, 2025

```
[1]: import numpy as np
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.inspection import PartialDependenceDisplay
import matplotlib.pyplot as plt

[2]: data = fetch_california_housing()

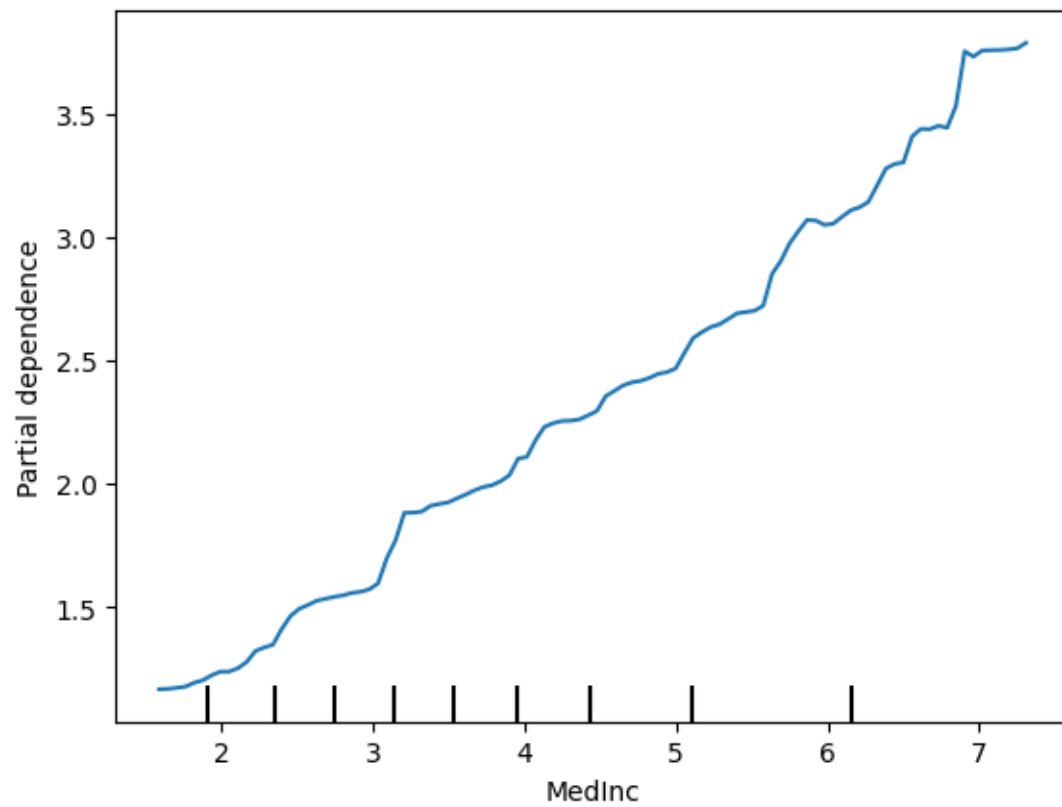
[3]: x = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target

[4]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)

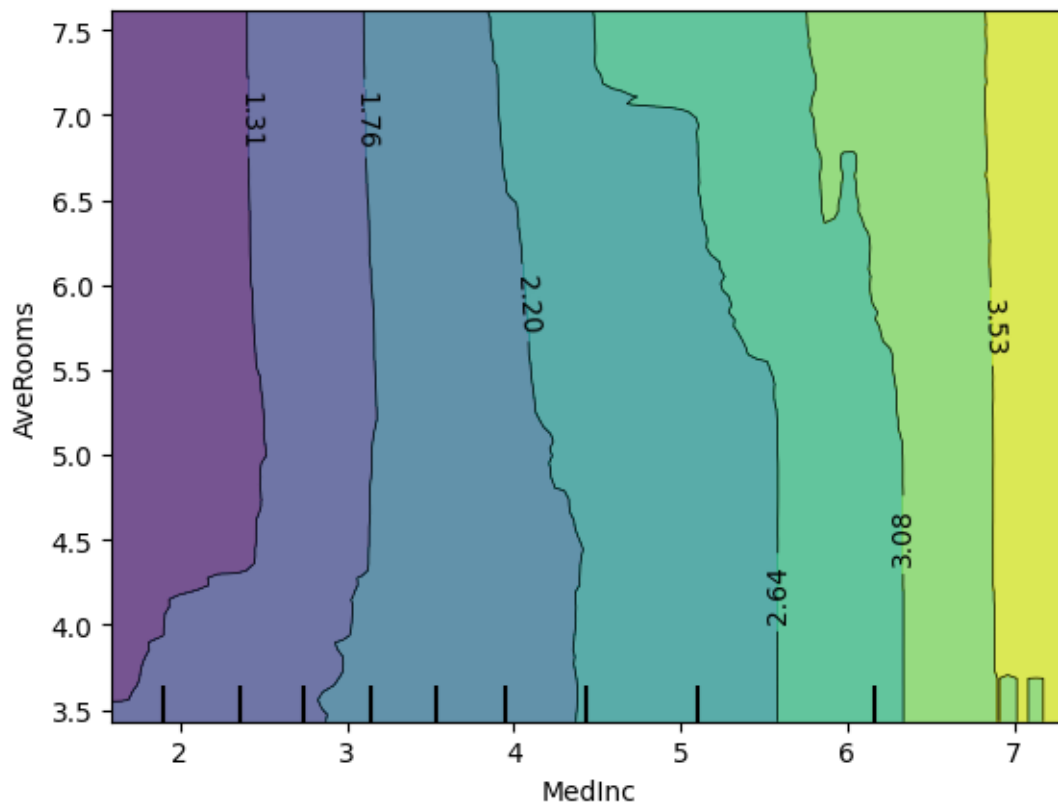
[5]: model = RandomForestRegressor(n_estimators=100)
model.fit(x_train, y_train)

[5]: RandomForestRegressor()

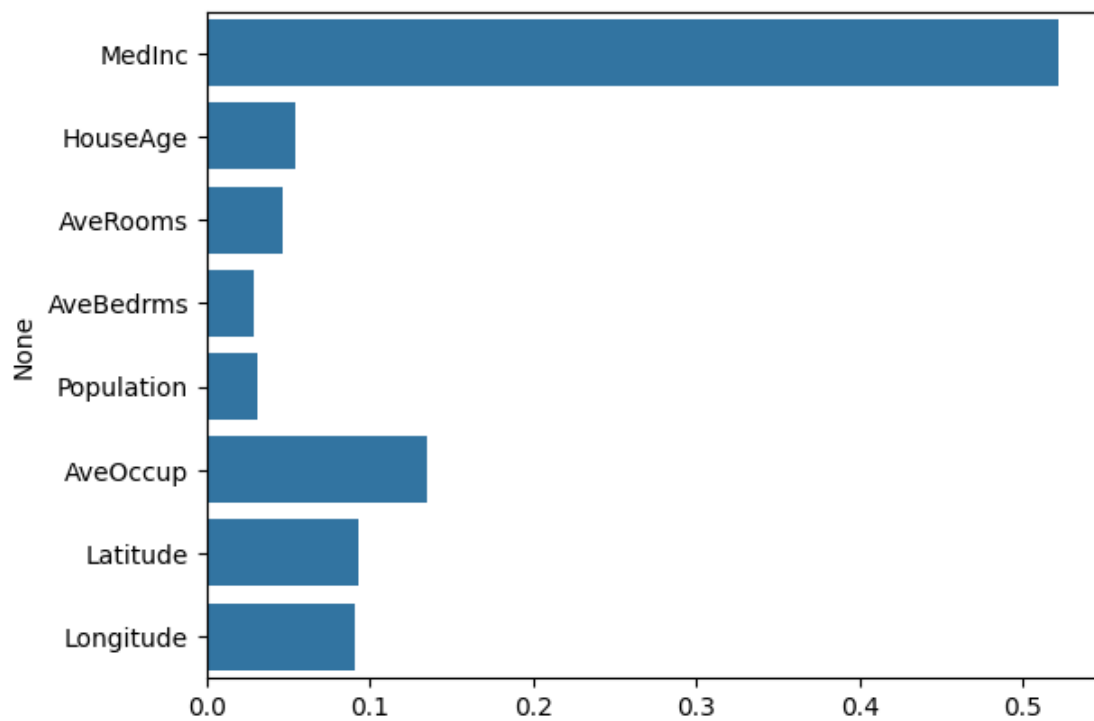
[6]: PartialDependenceDisplay.
↳ from_estimator(model, x_train, ["MedInc"], feature_names=data.feature_names)
plt.show()
```



```
[7]: PartialDependenceDisplay.  
     ↪from_estimator(model,x_train,[("MedInc","AveRooms")],feature_names=data.  
     ↪feature_names)  
plt.show()
```



```
[10]: imp_feat = model.feature_importances_
      fea = x_train.columns
      import seaborn as sns
      import matplotlib.pyplot as plt
      sns.barplot(x=imp_feat,y=fea)
      plt.show()
```



[]: